Loudspeakers and Headphones: An Assessment of Accuracies in Frequency Boost Identification by Trained Listeners

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ABSTRACT
Much research has been conducted within the industries centered around the construction of Headphones and Loudspeakers. While this has been the case, there has not been any research that has looked specifically at which environment audio engineers can better hear frequency boosts within objectively. Existing research has only looked at the issue based on how listeners compare in rating subjective qualities about the two spaces [1], [2]. To analyze this question from a purely objective point of view, we performed a common listening test consisting of three jazz samples between industry-standard pairs of headphones and loudspeakers. This listening test involved two groups of seven trained listeners (14 total) listening to the samples and deciding what frequency range was being boosted (9 DB boost, Q=2). To analyze the results of these tests, researchers will utilize a variety of statistical tests, including ANOVA (to test variance) and paired-sample t-tests (to test for significant differences). After the tests conclude, the researchers hope to discover a more concrete answer to this complicated question, which will have a major impact on the workflow of audio engineers at all levels of the profession.

1 Introduction

One of the more hotly debated issues in the audio engineering world today involves the question of what listening environment is the purest and most sonically representative; Loudspeakers or Headphones. This debate often occurs with people in many different specializations within the industry, whether mixing or mastering engineers, audio equipment designers, or researchers in general who uses listening tests for their research.

Every listening environment has positive and negative traits that draw listeners towards listening in these environments. Some claim that one "can hear even the most subtle details of a performance like the singer's breaths, fingers sliding over the strings or the recording venue's acoustics over a decent set of headphones." On the other hand, speakers are considered the most realistic listening environment. [3] Headphones are also considered the most flexible to use in various situations since they provide support in isolating the overall room environment. In addition, headphones are considered significantly cheaper as an investment than a brand-new pair of loudspeakers.

While people always seem to have subjective opinions on what they prefer, there is a surprising need for people who have thought to find out if there is a difference between what an audio engineer is listening to objectively in the two environments. This brought us to pose the crucial question; "In which environment can people most accurately detect boosts or cuts in specific frequency ranges? People have subjective opinions on what they like better, but in what environment are people the most objectively accurate? Is it worth investing in the more high-cost listening setups if it could be discovered that lesser expensive options provide similar enough results among audio engineers to convince an up-and-coming audio engineer to impact what product they decide to buy?

My research would test this question by performing the same standard listening test twice using various musical excerpts. This listening test would be tested with industry-standard studio headphones and an industry-standard loudspeaker setup. The results of this research would contribute to the field by providing objective data on whether there truly is a difference in the accuracy of frequency detection between these two different listening environments.

Are loudspeakers that much "better" than headphones (at least enough to warrant the additional expense for a professional listener or in the education process of a professional listener)?

2 Prior Art

Most comparative research that has been performed up to this point look at subjective qualities of preference between the two listening environments. While this has been the case, there has been some evidence contained within some studies to suggest that listeners may make more informed decisions "through headphones".. having "enabled better consistency between the listeners than the loudspeaker setups" [1].

While this conclusion was made in a work about subjective measurements, research concerning slightly more objective data collection strategies has yielded differing results. "Distributions when mixing over loudspeakers are markedly tighter, especially for rock and classical music" [2]. Tighter distributions in data suggest that listeners could more accurately hear the "correct change" more closely than those with looser distributions. Interestingly, such different conclusions were made in two separately drawn-out studies. This conversation within the context of musical genres provides an interesting wrinkle to consider when deciding when one environment is more accurate than the other. The study with R. King et al. says that their results conducted using a jazz excerpt led to the opposite effects ("showed a variance that was significantly higher for loudspeakers") compared to the other two genres that they tested [2]. In our research, we would like to explore this area further, and see how listeners would digest jazz samples in particular.

One important consideration that we think also has to be made concerns the conditions of the listening environment in the loudspeaker environment. While there are arguments for laying out these conditions in various ways, we think creating the best quality listening environment possible for the test is essential. This environment needs to be as close to the typical environment as most trained listeners/audio engineers would normally listen within. This is because "the modal artifacts introduced by the room can be so influential that they dominate the sound" [4] if the room is not designed properly with many of these issues in mind, that will lead to many problems.

A variety of factors (such as reverberation) can have an impact on the accuracy of the detection of musical characteristics. Research also suggests that the amount of reverberation in a room with loudspeakers may also impact the discrimination of specific musical characteristics (particularly with group delay) [5]. With all of these things in mind, it is important to note that "perceived differences exist in audio quality and musical performance between loudspeaker monitors and in-ear headphone monitors." [6]. While this research might not perfectly relate to the main topic at hand, this conclusion does clearly suggest the fact that auditory reflections coming from loudspeaker monitors likely does have a larger effect than we may even realize.

In our research, we intend to determine if there is a significant difference between the loudspeaker and headphone environments in the performance of everyday audio engineering tasks (identifying frequency boosts). In doing this, we intend to discover if extra reflections in an environment such as the loudspeaker environment cause additional problems for the everyday audio engineer trying to do their job.

3 Methods

Participants

14 graduate-level students (11 male and 3 female) were selected to participate in this study. The age range of the subjects in the study was from 22-30 years of age. All subjects reported having at least three months of extensive training as trained listeners.

The pool of 14 subjects was split into two randomized groups, each containing 7 participants. Each group was assigned a different test to start with (one group was assigned to start with the loudspeaker test, and the other group was assigned to start with the headphone test).

Tests

The two tests that were performed for this test are listed below. All the tests were graded strictly for objective accuracy (whether the participant identified the boost correctly based on the options that they were provided).

Loudspeaker Boost Identification Test
The Loudspeaker Boost Identification Test is a general audio engineering performative exam. This exam consisted of 45 multiple-choice questions and was administered within 23 minutes (30 seconds per question). The equipment that was utilized in this examination consisted of the technology found in the Belmont Listening Labs (JBL Speakers). This environment represents an "industry standard" setup that best represents the "typical" Loudspeaker mixing environment (not too cheap, not too expensive in price, the average environment). The playback volume of the samples being played was dictated based on the necessary industry testing standards in order to protect the hearing levels of the participants in the study.

Headphone Boost Identification Test
The Headphone Boost Identification Test is a general audio engineering performative exam. This exam consisted of 45 multiple-choice questions and was administered within 23 minutes (30 seconds per question). The equipment that was utilized in this examination consisted of the HD 280 Pro Studio Headphones. This headphone environment represents an "industry standard" setup that best represents the "typical" headphone listening environment (not too cheap, not too expensive in price, the average environment). Participants were allowed to adjust the volume to whatever volume they were most comfortable with to perform the test.

Stimuli
The stimuli used in this experiment consisted of 3 15-second samples (Samples A, B, and C) taken from various well-known jazz recordings. Each of the three samples were altered (based on boosts of 9 DB at 250 Hz, 500 Hz, 1K Hz, 2K Hz, and 4K Hz; Q=2). There were a total of 18 distinct audio samples created for the execution of the trials of this experiment (3 bypass examples, and 15 "question" examples with altered frequency information). The stimuli were designed to involve differences in instrumentation (duo, combo, jazz ensemble) and differences in frequency information contained in the audio samples. All stimuli were created by taking WAV files of the songs and cutting them into 15-second clips using the base equipment (the stock plugins and software) on Logic Pro.

Audio Sample	Boosts/Changes
Sample A (Round Midnight- Duo with Michael Brecker and Ray Brown, live performance from 2000)	Original track(unchanged), track boost at 250 Hz, 500 Hz, 1K Hz, 2K Hz, and 4K Hz
Sample B (Desafinado-Stan Getz and Joao Gilberto featuring Antonio Carlos Jobim)	Original track(unchanged), track boost at 250 Hz, 500 Hz, 1K Hz, 2K Hz, and 4K Hz
Sample C- (Count Basie-Corner Pocket, live performance in Stolkholm, 1962 featuring original Basie Band)	Original track(unchanged), track boost at 250 Hz, 500 Hz, 1K Hz, 2K Hz, and 4K Hz

Table 1. Breakdown of audio samples used for the various examinations. Each EQ boost outlined within the table consisted of a boost of 9 DB (centered at the frequency stated) with a Q factor of 2. The "Original Track" consisted of the original audio file with no change in EQ. This track was utilized as a "bypass" signal that the participants could use to go back and forth between in deciding their answers to each test question.

Procedure
Each test was conducted in person, ensuring to conform to the current ITB policy at Belmont University. When the participants entered the room to participate in the experiment, they were randomly split into two groups (of seven people). After being split up, the two groups were staggered in terms of which test they would perform first (Group 1 performed the Loudspeaker Test followed by the Headphone test; Group 2 performed the Headphone Test followed by the Loudspeaker Test). This decision was made to minimize the potential for a "training effect" as much as possible. Participants

were given a 10-minute break between the performance of the first and second tests.

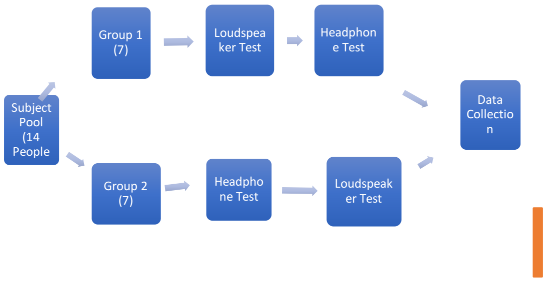

Figure 1. Breakdown of the pathway for the experimentation in terms of how the tests will be simultaneously performed (order). "Loudspeaker Test" and "Headphone Test" consist of the same number of questions ordered randomly for both tests.

Participants were allowed to cycle between the "question" (the boosted sample) as well as the original sample (the "bypass" button). Participants were given a time limit of 30 seconds for each question, where they were required to select their answer for what frequency range they felt was being boosted by the end of that time frame.

1. Question 1 *

   Mark only one oval.

   - ◯ 250 hz
   - ◯ 500 hz
   - ◯ 1000 hz
   - ◯ 2000 hz
   - ◯ 4,000 hz

Figure 2. Sample Question of what the participant filled out for each question within the examination. Participants will be played the question sample and can go between that and the "bypass" feature. Based on what frequency the participant believes is adjusted, the participant marked that oval within the online google form.

Results

The results of the experiment involved ninety test scores from each participant (45 questions per test, 2 tests in total). In total, there were 1260 individual data points (based on questions answered by the participants). To see the effects of the listening environment on the overall performance level of the subjects of the listening test, we will run an ANOVA based on the listening stimuli vs. the listening environment. After finding out if there is a significant variance between the main two variables of the hypothesis, we will also run a further ANOVA test to account for the additional variables of attenuated frequencies (250 Hz, 500 Hz, 1K Hz, 2K Hz, 4K Hz) and group variables (Group 1 and 2).

After running the ANOVA tests, we will run a variety of paired samples t-tests to gain further more detailed information on the data by comparing the means of the data (as opposed to just variance). In doing this, we will look at questions connecting to variables such as group variable (did group 1 or group 2 perform differently), attenuated frequencies (what frequency boosts were participants more accurate in hearing compared to others; i.e., 500 Hz vs. 4K Hz, 250 Hz vs. 500 Hz), listening stimuli (Sample A vs. B vs. C), and listening environment (Loudspeakers vs. Headphones).

If our hypothesis is correct, there will be a significant difference in how trained listeners perform in Headphones vs. Loudspeakers. Our worst-case scenario for this test would be that there is no significant difference between the two listening environments (we can't reject the null hypothesis).

4 Discussion

Based on the research that has been done so far, there is still a lot to be learned about the comparative performance of performing basic audio engineering tasks between the listening environments of Loudspeakers and Headphones. While this study only looked at answering the question based on the confines of one genre of music (Jazz), every genre of music has its own challenges that could impact how participants would perform in a listening test.

In addition, many other products not tested in this research could also be considered “industry standards” within the headphone and loudspeaker industries. Since the cost models of the industries are different in structure, it can be difficult to find the perfect point of comparison to compare these two environments. While this is true, we feel that it is more important to consider these two products as “listening environments” as opposed to just comparing the two environments based on aspects such as price point. There is plenty of research to compare different headphones or loudspeakers that didn’t have the same price point. We believe that this research should be considered in the same light as that research.

Previous research has yet to be particularly expansive on bridging this gap in comparing these two significant industries (Loudspeakers vs. Headphones). While it is difficult to execute such research properly, it is still essential, considering how crucial this question is for people at all levels of the industry. In addition, there has been no research performed previously looking at analyzing the differences in these two differing listening environments based on objective listening performance standards. Our study aims to find an initial answer to this crucial debate.

5 Conclusions

Our research aimed to assess the differences in effectiveness in the performance of basic tasks of audio engineering using headphones vs. loudspeakers in the jazz genre. With the execution of a common listening test in both environments using industry-standard equipment, we can gain a much fuller understanding of how the human ear functions differently within these two sonic environments. Even though our research limits itself to only one genre of music(jazz), our stimuli were expressly designed to provide as much variety as possible in terms of instrumentation and frequency information within the samples. Our hypothesis before executing this study is that there will be a significant difference in the effectiveness rates between the two environments.

Suppose we cannot reject the null hypothesis. In that case, the information that will still be provided with that result will still prove significant in aiming to help the impacted industries.

6 Further Research

Further research that could be performed in this area could include expanding the number of genres that are covered in this research. This would help widen the scope and the amount of data that could be collected to help provide a firmer understanding of this subject.

In addition, further research could be done in this light on the ability to perform other common tasks in audio engineering. This could include research such as studying delay identification or seeing if people can hear reverberation better between the two (among others). There is a lack of research in this area with objective data collection, so it could easily be argued that there are many paths that could be forged ahead after the result of this particular research.

References

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